



Evaluation Tasks

Evaluation type: Unsupervised remote evaluation
Estimated time: 30-45 minutes

1. Before Starting

In this evaluation, you will use MEDIATA to inspect several sample datasets, generate the metadata needed for integration, create or review mappings, process the datasets, and inspect the produced files.

The goal is to evaluate the tool workflow. You are not being evaluated on clinical knowledge nor federated learning. Focus on whether MEDIATA helps you find files, understand columns and values, build mappings, review suggestions, process data, and recover from mistakes.

Before starting:

- Launch the local deployment following the [readme.md local deployment section](#) from the [umbrella repository](#).
- Open the tutorial page from the navigation bar in a new tab.
- Keep this task sheet and the questionnaire available while working.

2. Evaluation Focus

These tasks evaluate three parts of MEDIATA:

- Node and Metadata: initial user interaction with the interface.
- Discovery: finding datasets, selecting several files, inspecting columns, values, data types, filters, aggregate statistics, and data-quality signals.
- Integration: generating element metadata, loading those element files, using or creating mappings, reviewing suggested mappings, loading a schema or mapping specification, processing datasets, and checking the resulting files.

When a task asks you to "record" something, write it in the questionnaire.

3. Node and Metadata Tasks

Task 0 - Open the Local Node and FAIR Data Point

Purpose: Check whether users can reach the local workspace and understand where public metadata is exposed.

1. Log in with admin / admin.
2. Select the default project.
3. Open the MEDIATA Node.
4. Open the Node metadata panel.
5. Inspect the dataset descriptions shown in the metadata panel.
6. Find the FAIR Data Point badge.
7. Copy the FAIR Data Point URL and open it in another browser tab.

Record whether the project, Node, metadata panel, dataset descriptions, and FAIR Data Point URL were easy to find and understand.

4. Discovery Tasks

Task 1 - Open Discovery and Load Sample Datasets

Purpose: Check whether users can find the data inspection area and load several datasets.

1. Open the Discovery tab. The File Explorer will open.
2. Open sample_dataset_1.csv first.
3. Reload the page, then open all four sample datasets together.

Use multi-selection when needed: Ctrl-click or Shift-click on desktop, or long-press a file row to enter multiple-selection mode.

Record whether file selection, multi-selection, and opening files were clear.

Task 2 - Select Files and Hide One Feature

Purpose: Check whether users understand that loaded files, selected files, and visible features are controlled separately.

1. Use the Discovery file selector to enable all loaded sample datasets.
2. In the feature toggles, search for fim_total.
3. Hide fim_total from the current view.
4. Confirm that it disappears from the tables/charts.

5. Show fim_total again before continuing.

Record whether it was clear how to select files, search for a feature, hide it, and show it again.

Task 3 - Inspect Columns and Value Summaries

Purpose: Check whether users can inspect the information needed before integration.

1. Find one sex field and one smoking-related field.
2. Find one diagnosis field.
3. Find at least one functional score column, such as an eating, bathing, toileting, FIM, or Barthel field.
4. Inspect the displayed type, examples, counts, charts, and missing-value information.
5. If a feature seems to be shown in the wrong type table, use the selected feature type switch to view it as categorical or continuous.

Record whether equivalent fields across files were easy to recognize, and whether column names, values, data types, and summaries were understandable.

Task 4 - Filter One Dataset

Purpose: Check whether users can focus on a subset of records without manually scanning the whole table.

1. In Discovery, show only sample_dataset_1.csv.
2. Add filters to show male patients with a FIM total under 15, or the closest available fields if names differ.
3. Observe whether the charts and tables change.
4. Reset the filters.
5. Enable all sample files again.

Record whether the filter controls and the updated charts were clear.

Task 5 - Review Aggregate Metrics

Purpose: Check whether users can inspect relationships between columns and understand omitted features.

1. Switch to aggregate metrics.
2. View covariance.
3. Change the metric to Pearson correlation.
4. Change the metric to Spearman correlation.
5. Resize the upper and lower panels.
6. Inspect the omitted-features panel, if any omitted features are listed.

Record whether the metric selector, resizing, and omitted-feature UI were understandable.

Task 6 - Detect and Clean a Planted Outlier

Purpose: Check whether users can notice an obvious data-quality issue and apply a concrete cleaning step.

1. In Discovery, inspect sample_dataset_1.csv.
2. Select a continuous measurement field and use the integrity metrics or outlier indicators to identify an obvious outlier.
3. Reload Discovery, in the File Explorer, select sample_dataset_1.csv, and open Data cleaning.
4. Choose Remove rows with pattern.
5. Configure it for the column containing the outlier with a pattern that removes only the anomalous row you identified. For example, use an exact-match regular expression such as ^<value you saw>\$ rather than a broad pattern that would remove normal rows.
6. Apply the cleaning step.
7. Open the cleaned output in Discovery and confirm that the outlier is gone.

Record whether the outlier was visible, whether the cleaning operation was understandable, and whether the cleaned output was easy to verify.

Task 7 - Generate Element Metadata for Integration

Purpose: Check whether users can move from Discovery into Integration using the intended tool flow.

1. Return to Discovery with all four sample datasets loaded.
2. In the Discovery file selector, make sure the sample datasets you want to use are enabled and visible in the current view.
3. Use Upload elements to generate/upload element metadata for the datasets currently shown in Discovery.
4. Note the generated element files and where they appear.

Record whether it was clear that element generation depends on which datasets are enabled in Discovery, and whether Integration uses generated element metadata rather than the raw datasets directly.

5. Integration Tasks

Task 8 - Open Integration and Load Element Files

Purpose: Check whether users can start the mapping workflow with all relevant files.

1. Open the Integration tab.
2. In the File Explorer, select the generated element files for all four sample datasets.
3. Use Ctrl-click, Shift-click, or long-press to select multiple files.
4. Open them and inspect the Integration screen.

Record whether the feature list, selected feature area, mapping controls, and resulting mapping panel were clear.

Task 9 - Create a Simple Manual Smoking-History Mapping

Purpose: Check whether users can create a small categorical mapping by hand for a realistic harmonization case.

1. Select the available smoking-related columns.
2. Create a target mapping such as `smoking_history`.
3. Add normalized values such as Never, Former, Current, and Unknown.
4. Assign the source categories to the normalized output values.
5. Add the mapping to the result.

Record whether value selection and categorical mapping were understandable.

Task 10 - Load the Target Schema and Rename Bathing

Purpose: Check whether users can load a schema and use it to guide mapping work.

Use `evaluation/sample_schemas/sample_schema.json`.

1. Open the Schema tab.
2. Upload the target schema.
3. Inspect the target fields.
4. Use the schema-guided suggestions to check whether the loaded fields now suggest the expected schema targets.
5. Find or create/review the bathing-related mapping.
6. Using the in-app mapping editor, rename the resulting bathing column to `bathing_status`.
7. Confirm that the renamed mapping appears in the column name suggestions as expected.

Record whether the schema loaded, whether schema-guided suggestions pointed to expected target fields, and whether renaming the bathing mapping in the editor was clear.

Task 11 - Run Suggested Mappings

Purpose: Check whether users can generate, inspect, and correct automatic mapping suggestions.

1. Click Suggest mappings.
2. Use the option that replaces the current result.
3. Inspect the suggested column groups.
4. Inspect suggested values or numeric ranges where they are shown.
5. Correct one questionable mapping if you find one.

Pay special attention to whether similar concepts are grouped sensibly, such as sex with sex, smoking history with smoking history, eating with eating, bathing with bathing, and toileting with toileting.

Record whether suggestions were useful, whether the grouped concepts made sense, and what you corrected, if anything.

Task 12 - Upload the Sample Mapping Specification

Purpose: Check whether users can load a prepared mapping file instead of building everything from scratch.

Use `evaluation/sample_mappings/sample_spec.json`.

1. Upload the sample mapping specification.
2. Resolve source files if the interface asks for it.
3. Confirm that mappings appear in the result panel.

Record whether upload, file resolution, and loaded mappings were understandable.

Task 13 - Edit One Existing Mapping

Purpose: Check whether users can safely adjust the mapping specification.

Choose one mapping to inspect or modify. Suggested options:

- `smoking_history`
- `bmi`
- an eating, bathing, toileting, FIM, or Barthel score mapping

Record whether you could inspect or edit the mapping successfully, and whether the edit controls were clear.

Task 14 - Process the Datasets

Purpose: Check whether users can apply the mapping to produce harmonized output files.

1. Click Process datasets.
2. In the processing modal, select each source dataset that should be transformed.
3. Check that the modal and listed files make sense.
4. Apply the mapping.

Record whether it was clear that each file needed to be selected and whether the processing modal was understandable.

Task 15 - Save, Export, and Reopen the Results in Discovery

Purpose: Check whether users can preserve the mapping work and validate the produced files.

1. Save or download the mapping specification.
2. Return to the File Explorer.
3. Find the produced parsed_ files.
4. Rename the produced files if needed.
5. Open the produced files in Discovery.
6. Inspect their shape, columns, and at least one value summary.
7. Use the feature search/toggles to focus on one resulting score column and one resulting categorical column.

Record whether the output files were easy to find and rename, and whether the parsed file shape, columns, and value summaries were expected.

6. After Completing the Tasks

After finishing the workflow:

- Fill in the questionnaire.
- Mark each task as completed, partially completed, or not completed.
- Rate the difficulty of each task.
- Complete the usability statements.
- Add comments where something was confusing, unclear, or unexpected.
- Submit the completed questionnaire using the method provided with the release package.